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Cost-Effectiveness of ApoB, Non-HDL-C, and LDL-C Goals for Primary Prevention Lipid-Lowering Therapy

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IMPORTANCE Apolipoprotein B (apoB) is a superior marker of residual atherosclerotic cardiovascular disease risk in patients treated with lipid-lowering therapy (LLT) compared with low-density lipoprotein cholesterol (LDL-C) and non-high-density lipoprotein cholesterol (non-HDL-C). The cost-effectiveness of LDL-C, non-HDL-C, and apoB goals has not been established.

OBJECTIVE To determine the relative cost-effectiveness of intensifying LLT for primary prevention based on LDL-C, non-HDL-C, and apoB goals.

DESIGN, SETTING, AND PARTICIPANTS This economic evaluation used a computer simulation model to evaluate the cost-effectiveness of intensifying LLT with high-intensity statins or ezetimibe according to LDL-C, non-HDL-C, or apoB goals. A cohort of 250 000 statin-eligible and atherosclerotic cardiovascular disease-free US adults was constructed from 2005 to 2016 National Health and Nutrition Examination Survey participants (N = 4149). Individuals commenced the simulation after lipid screening and received statin therapy based on 2018 American Heart Association/American College of Cardiology guidelines. Model inputs were derived from national survey data, pooled longitudinal cohort studies, and published literature. Uncertainty was explored with traditional and probabilistic sensitivity analysis.

EXPOSURES Lipid-lowering therapy was intensified if individuals did not achieve treated LDL-C level less than 100 mg/dL, non-HDL-C level less than 118 mg/dL, or apoB level less than 78.7 mg/dL.

MAIN OUTCOMES AND MEASURES Lifetime quality-adjusted life-years (QALYs) and costs (in 2025 US dollars), discounted 3.0% annually. The primary outcome was the incremental cost-effectiveness ratio. Strategies were considered cost-effective if they cost less than \$120 000 per QALY gained.

RESULTS Compared with an LDL-C goal, 965 QALYs (95% uncertainty interval [UI], –3551 to 5341 QALYs) would be gained with a non-HDL-C goal, alongside a \$2.1 million (95% UI, –\$94.2 million to \$92.0 million) reduction in costs. Compared with a non-HDL-C goal, 1324 QALYs (95% UI, –2602 to 5669 QALYs) would be gained with an apoB goal, alongside a \$40.2 million (95% UI, –\$43.6 million to \$134 million) increase in costs, yielding an incremental cost-effectiveness ratio of \$30 300 per QALY gained. At a willingness-to-pay threshold of \$120 000 per QALY gained, an apoB goal was optimal in 65% of probabilistic analyses and a non-HDL-C goal was optimal in 25%. The cost of apoB testing was marginal; higher costs reflected longer life expectancy and prolonged preventive treatment.

CONCLUSIONS AND RELEVANCE The results of this computer simulation study suggest that apoB can be used as a cost-effective marker to guide primary prevention LLT and improve population health.

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 Editorial

 Supplemental content

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Atherosclerotic cardiovascular disease (ASCVD) events have a substantial economic effect, with projected costs exceeding \$1 trillion by 2035.¹ Serum concentrations of apolipoprotein B (apoB)-containing lipoproteins are the primary causal determinants of ASCVD events.² These lipoproteins include low-density lipoprotein cholesterol (LDL-C), remnants, very LDL-C, intermediate-density lipoprotein cholesterol, and lipoprotein a. Pharmacologic reduction of atherogenic lipid concentrations in at-risk patients is a cornerstone of ASCVD primary prevention strategies.

The adequacy of atherogenic lipid-lowering therapy (LLT) can be assessed through the measurement of on-treatment LDL-C, non-high-density lipoprotein cholesterol (non-HDL-C), and apoB serum concentrations.³ Low-density lipoprotein cholesterol and non-HDL-C reflect the mass of cholesterol within atherogenic fractions, whereas apoB directly measures the concentrations of all atherogenic particles in the blood. According to results from observational studies, mendelian randomization studies, and randomized clinical trials, apoB has been determined to be a more accurate marker of ASCVD risk than LDL-C or non-HDL-C.^{4,5} Multiple post hoc analyses of LLT clinical trials have shown that on-treatment change in apoB serum concentrations associates more strongly with ASCVD risk than on-treatment change in non-HDL-C and LDL-C, suggesting that lower apoB level may be a superior therapeutic target for patients undergoing LLT.⁶⁻⁸

Despite strong evidence that apoB is a superior marker of ASCVD risk, clinical guidelines largely rely on LDL-C levels to guide decision-making regarding initiation and intensification of LLT.³ Specifically, American Heart Association/American College of Cardiology (AHA/ACC) 2018 guidelines suggest that LLT should be intensified if patients fail to achieve LDL-C level less than 100 mg/dL. Although measuring apoB adds testing costs to traditional lipid panel expenditures, this up-front expense might be justified by the benefit of more accurately allocating intensive LLT to individuals most likely to benefit from treatment, which could reduce downstream ASCVD events and attributable health care costs. To assess the value of available LLT goals, we conducted a computer simulation study to estimate the cost-effectiveness of LDL-C, non-HDL-C, and apoB goals for the intensification of primary prevention LLT.

Methods

We simulated 3 treatment strategies using the CVD Policy Model (eTables 1 and 2 and eFigures 1 and 2 in [Supplement 1](#)), a previously published computer simulation model.^{9,10} All outcomes were presented using the mean and 95% uncertainty interval (UI) from 1000 probabilistic analyses, in which model parameters were sampled from predetermined statistical distributions ([Table 1](#)).¹¹ We used a health sector perspective and adhered to Consolidated Health Economic Evaluation Reporting Standards (CHEERS) guidelines (eTable 3 in [Supplement 1](#)).¹² Institutional review board approval was not required because this economic evaluation study was a secondary analysis of publicly available and deidentified data. To convert cholesterol values to mmol/L throughout, multiply by 0.0259.

Key Points

Question In primary prevention, would using an apolipoprotein B (apoB) goal to guide intensification of lipid-lowering therapy (LLT) be more cost-effective than a low-density lipoprotein cholesterol (LDL-C) or non-high-density lipoprotein cholesterol (non-HDL-C) goal?

Findings In this computer simulation economic evaluation study, the cost-effectiveness of LDL-C, non-HDL-C, and apoB goals for LLT was compared in a cohort of 250 000 atherosclerotic cardiovascular disease-free and statin-eligible US adults. An apoB goal was estimated to produce the most quality-adjusted life-years and was highly cost-effective.

Meaning The results of this study support the use of apoB as a cost-effective marker to guide primary prevention LLT and improve population health.

Table 1. Baseline Descriptive Statistics of the Cohort (N = 4149)^a

Characteristic	No. (%)
Age, mean (SD), y	66.5 (11.0)
Men	2458 (59.2)
Women	1691 (40.8)
Race and ethnicity ^b	
Non-Hispanic Black	894 (21.5)
Mexican American	876 (21.1)
Non-Hispanic White	2083 (50.2)
Other race, including multiracial	296 (7.1)
Diabetes	1433 (34.5)
Blood pressure medication use at baseline	2531 (61.0)
Estimated 10-y ASCVD risk, mean (SD), % ^c	20.9 (14.2)
Body mass index, mean (SD)	29.5 (6.4)
Systolic blood pressure, mean (SD), mm Hg	132.0 (18.4)
Diastolic blood pressure, mean (SD), mm Hg	69.1 (12.5)
Currently smokes	860 (20.7)
Formerly smoked	1432 (34.5)
Low-density lipoprotein cholesterol, mean (SD), mg/dL	119.2 (42.2)
Non-high-density lipoprotein cholesterol, mean (SD), mg/dL	166.5 (45.9)
Apolipoprotein B, mean (SD), mg/dL	110.8 (39.0)
High-density lipoprotein cholesterol, mean (SD), mg/dL	53.3 (16.6)

Abbreviation: ASCVD, atherosclerotic cardiovascular disease. Body mass index is calculated as weight in kilograms divided by height in meters squared.

SI conversion factor: To convert cholesterol to mmol/L, multiply by 0.0259; to convert apolipoprotein B to g/L, multiply by 0.01.

^a Statin-eligible individuals from the 2005 to 2016 National Health and Nutrition Examination Survey (NHANES) were probabilistically sampled with replacement to create a simulation cohort of 250 000.

^b Race and ethnicity were categorized according to NHANES classifications.

^c Estimated with the pooled cohorts ASCVD risk equation.¹¹

CVD Policy Model

The CVD Policy Model estimated lifetime health-related quality of life and direct health care costs for lipid-lowering treatment strategies in a nationally representative cohort of ASCVD-free US adults, which has been described extensively previously.⁹ This model uses time-varying ASCVD risk factor

exposures to simulate an individual's lifetime risk of ASCVD. As individuals progress through the model, they accumulate health care costs and quality-adjusted life-years (QALYs), with these values derived from published literature and publicly available data. The model performs individual-level simulation to predict lifetime risks and health and cost consequences of ASCVD events. Individuals enter a simulation with no established ASCVD and face annual risks of coronary heart disease, stroke, and non-ASCVD mortality. Risk of events is predicated on patients' risk factor profiles and logistic risk functions developed in the National Heart, Lung, and Blood Institute Pooled Cohorts Study (NHLBI-PCS) dataset.¹³ The risk functions were recalibrated to match ASCVD-related mortality and all-cause mortality rates in the wider US population (eFigure 2 in Supplement 1).⁹

Simulation Cohort

The CVD Policy Model requires a simulation cohort, composed of individuals with complete risk factor information for every year in the study time horizon. Risk factors included in the model are age, sex, race and ethnicity, LDL-C level, HDL-C level, systolic blood pressure, diastolic blood pressure, body mass index (calculated as weight in kilograms divided by height in meters squared), smoking status, estimated glomerular filtration rate, and diabetes status. We derived our simulation cohort from ASCVD-free participants from the 2005 to 2016 examinations of the US National Health and Nutrition Examination Survey (NHANES).¹⁴ These examinations were selected because they included apoB measurement. Trajectories for most ASCVD risk factors were assigned to NHANES participants by randomly matching them 1:1 to NHLBI-PCS participants for whom trajectories were previously defined.⁹ Because apoB was not measured in all NHLBI-PCS studies, apoB values were imputed by assuming that patients' LDL-C to apoB ratio was consistent throughout their lifetime. This assumption preserved discordance between these markers over time. To validate this assumption, we analyzed the 2005 to 2016 NHANES examinations and demonstrated that the mean ratio of LDL-C to apoB is consistent across 4 decades of age (40-80 years) for individuals with and without diabetes, with narrow SDs ranging from 0.16 to 0.19 (eTable 4 in Supplement 1).

Individuals were included in the simulation cohort if they were eligible for statin therapy according to the AHA/ACC guidelines.³ Specifically, individuals were included if they had no established ASCVD, baseline estimated 10-year ASCVD risk greater than or equal to 7.5%, established diabetes, LDL-C level greater than or equal to 190 mg/dL, or borderline risk (10-year ASCVD risk, 5.0%-7.5%) with untreated LDL-C level greater than or equal to 160 mg/dL. After inclusion criteria were applied, 4149 NHANES participants remained (eFigure 3 in Supplement 1). We probabilistically sampled these individuals with replacement to create a cohort of 250 000 individuals for analysis.

Treatment Strategies

Individuals commenced the simulation by presenting to a primary care physician for ASCVD risk screening and received a statin therapy prescription based on AHA/ACC guidelines.³ Mod-

erate-intensity statin therapy was initiated for individuals with a 10-year estimated ASCVD risk of at least 7.5%, borderline risk with LDL-C level greater than or equal to 160 mg/dL or non-HDL-C level greater than or equal to 190 mg/dL, or diabetes. High-intensity statins were initiated for individuals with elevated ASCVD risk and diabetes. Individuals who developed diabetes during follow-up switched to high-intensity statins.

We simulated 3 alternative treatment goals for each individual. Routine blood tests were conducted for each simulated individual 3 months after treatment initiation. In the usual care group, patients with treated LDL-C level greater than or equal to 100 mg/dL had their treatment intensified. Treatment for patients above goal and receiving initial moderate-intensity statins was intensified to high-intensity statins. Patients still remaining above goal and receiving initial high-intensity statins had treatment intensified to receive supplemental ezetimibe therapy. In the non-HDL-C goal group, patients with treated non-HDL-C level greater than or equal to 118 mg/dL were considered above goal and experienced similar treatment intensification steps. In the apoB goal group, patients with treated apoB level greater than or equal to 78.7 mg/dL were considered above goal and had treatment intensified similarly. Non-high-density lipoprotein cholesterol and apoB treatment goals were selected because they were percentile equivalents to the guideline-recommended target of LDL-C at 100 mg/dL. Because statins reduce cholesterol mass more efficiently than they reduce atherogenic particle number,⁸ this approach led to more treatment intensification for patients under non-HDL-C and apoB strategies than LDL-C. In selecting percentile-equivalent thresholds in the untreated population, we interpreted the higher level of treated apoB compared with LDL-C with any unit dose of treatment as a signal that further benefit can be achieved with apoB-guided care. A sensitivity analysis, described below, considered the cost-effectiveness of intensifying treatment for a fixed number of individuals with different LDL-C, non-HDL-C, and apoB goals.

Simulating Treatment

Mean (SD) reductions in LDL-C, non-HDL-C, and apoB levels from moderate-intensity statins were 30.4% (6.8%), 28.0% (6.4%), and 23.7% (5.3%), respectively, derived from primary prevention clinical trials that reported reduction in all 3 lipoproteins (eTable 5 in Supplement 1).⁸ For high-intensity statins, mean (SD) reductions in LDL-C, non-HDL-C, and apoB levels were 43.1% (2.7%), 38.7% (2.5%), and 32.4% (2.1%), respectively. When ezetimibe was added to statin therapy, additional mean (SD) reductions in LDL-C, non-HDL-C, and apoB levels were 25.0% (2.2%), 23.0% (1.7%), and 18.9% (1.6%), respectively.¹⁵ We modeled the lipid-lowering effects for each lipoprotein by using normal distributions. In each probabilistic iteration of the analysis, a single percentile was sampled and applied across all 3 distributions, ensuring perfect rank correlation in mean lipid reductions. Within each probabilistic iteration of the model, 250 000 patients were simulated. For each patient, we represented heterogeneity in individual treatment response by assigning each individual a lipid reduction drawn from a patient-level distribution centered on the iteration-specific mean reduction. The variance of this distribution was

derived from a meta-analysis.¹⁶ This approach captures uncertainty in average treatment effects across simulations; expected ordering of reductions in individuals' LDL-C, non-HDL-C, and apoB levels; and individual variability (random variation around the mean) in response to treatment.

Reductions in lipoproteins reduced patients' risk of incident coronary heart disease (relative risk per reduction in apoB level of 10 mg/dL, 0.90; 95% CI, 0.84-0.96) and incident stroke (relative risk per reduction in apoB level of 10 mg/dL, 0.93; 95% CI, 0.86-1.02) (eTable 5 in [Supplement 1](#)).¹⁷ To validate that parameterizing treatment effects through apoB level reduction did not artifactually inflate statin benefits, we simulated moderate- and high-intensity statin therapy during 5 years and found implied relative risk of 0.80 and 0.81 per LDL-C reduction of 1.0 mmol/L, closely matching results from a large meta-analysis of statin benefit with median follow-up of 5.1 years (eTable 6 in [Supplement 1](#)).¹⁸

Annual prices for statin therapy and ezetimibe were obtained from the median price of these medications in the Medical Expenditure Panel Survey 2022, alongside a dispensing fee (eTable 5 in [Supplement 1](#)).^{9,19} All individuals experienced pill-taking disutility and individuals receiving statin therapy faced a risk of statin-induced diabetes, minor adverse events, and major adverse events, with effects on cost and quality of life.⁹ Individuals commencing high-intensity statin therapy experienced a relative risk of 1.12 (95% CI, 1.04-1.22) for statin-induced diabetes compared with moderate-intensity statin therapy.

Outcomes

Primary outcomes were direct health care costs (in 2025 US dollars), QALYs, and the incremental cost-effectiveness ratio (ICER). Future costs and QALYs were discounted 3.0% annually.²⁰ Any costs sourced in years before 2025 were inflated using the US Bureau of Labor Statistics Consumer Price Index for medical care.²¹ We used a health sector perspective, capturing direct medical costs relevant to health care decision-makers, and a lifetime horizon. A strategy was considered cost-effective if it cost less than \$120 000 per QALY, as recommended by the AHA/ACC.²² Secondary outcomes recorded were ASCVD events prevented, life-years gained, and disaggregated health and cost outcomes.

Statistical Analysis

We conducted an extensive 1-way sensitivity analysis to examine the effect of key modeling assumptions on predicted outcomes. Results from this analysis were collated in a tornado diagram. To demonstrate that incremental gains from non-HDL-C and apoB goals are not simply due to intensifying treatment for more individuals, we examined the effect of using LDL-C, non-HDL-C, and apoB goals that intensified treatment for approximately the same number of patients as LDL-C level greater than or equal to 100 mg/dL and apoB level greater than or equal to 78.7 mg/dL, respectively. Finally, we conducted the analysis in sex-disaggregated cohorts of 250 000 ASCVD-free individuals who met analysis inclusion criteria. Analyses were conducted from January 10, 2026, to February 20, 2026, in TreeAgePro version 2024 (TreeAge Software).

Results

Base Case

The combined cohort consisted of 250 000 statin-eligible and ASCVD-free individuals (2458 men [59%]; 1691 women [40.8%]; mean [SD] age, 66.5 [11.0] years; mean [SD] LDL-C level, 119.2 [42.2] mg/dL; mean [SD] apoB level, 110.8 [39.0] mg/dL; mean [SD] ASCVD 10-year risk score, 20.9% [14.2%]) probabilistically sampled from 4149 participants from 2005 to 2016 examinations of NHANES (Table 1). The weighted cohort was racially and ethnically diverse and categorized according to NHANES classifications: 894 (21.5%) were non-Hispanic Black; 876 (21.1%), Mexican American; 2083 (50.2%), non-Hispanic White; and 296 (7.1%), of other or multiracial backgrounds. Men had a lower mean (SD) age than women in the cohort (64.0 [10.6] vs 69.9 [10.5] years), lower mean (SD) LDL-C level (116 [41] vs 123 [44] mg/dL), and lower mean (SD) apoB level (108 [35] vs 115 [43] mg/dL) (eTable 7 in [Supplement 1](#)).

Compared with usual care (LDL-C goal of 100 mg/dL), intensifying LLT based on a non-HDL-C goal was estimated to prevent 617 ASCVD events (95% UI, -245 to 1422) per 250 000 individuals initiating statin therapy (Table 2; eFigure 4 in [Supplement 1](#)). This intensification strategy was estimated to be cost saving compared with usual care, indicating reduced costs and improved population health. A non-HDL-C goal would produce approximately 965 QALYs (95% UI, -3551 to 5341 QALYs) and result in an approximately \$2.1 million (95% UI, -\$94.2 million to \$92.0 million) reduction in costs.

Compared with a non-HDL-C goal of 118 mg/dL, intensifying LLT based on an apoB goal was estimated to prevent 1018 ASCVD events (95% UI, -1974 to -6 ASCVD events) per 250 000 individuals initiating statin therapy, producing 1324 QALYs (95% UI, -2602 to 5669 QALYs) and costing approximately \$40.2 million (95% UI, -\$43.6 million to \$134 million) (Table 2; eFigure 4 in [Supplement 1](#)). Incremental to the non-HDL-C strategy, apoB-guided intensification would be cost-effective (ICER, \$30 300 per QALY gained).

At a willingness-to-pay threshold of \$120 000 per QALY gained, a non-HDL-C goal was optimal in 25% of probabilistic iterations and an apoB goal was optimal in 65% of probabilistic iterations of the model (Figure 1). Compared with an LDL-C goal, non-HDL-guided intensification was estimated to have an ICER of \$105 per QALY for women and was cost saving for men (eTables 8 and 9 and eFigures 5 and 6 in [Supplement 1](#)). Compared with a non-HDL-C goal, apoB-guided intensification had an ICER of \$35 900 per QALY for women and \$26 600 per QALY for men.

Compared with usual care, both non-HDL-C and apoB goals were predicted to modestly reduce lifetime chronic and acute ASCVD costs but would increase statin therapy and non-ASCVD expenditure (Table 3; eTable 10 in [Supplement 1](#)). The contribution of apoB testing costs to overall incremental costs was small compared with broader health care spending; hence, the increase in costs in the apoB goal group was largely due to extension of life expectancy and extended years of preventive treatment.

Table 2. Projected Outcomes for LDL-C-, Non-HDL-C-, and ApoB-Guided Treatment Intensification for 250 000 Simulated Individuals^a

	Mean (95% UI) ^b				
	LDL-C goal (100 mg/dL) usual care	Non-HDL-C goal (118 mg/dL)	ApoB goal (78.7 mg/dL)	Non-HDL-C goal vs LDL-C goal	ApoB goal vs non-HDL-C goal
Primary outcomes					
Cost, thousands (in 2025 dollars)	58 062 120 (57 648 109 to 58 471 283)	58 059 999 (57 645 455 to 58 461 278)	58 100 177 (57 681 892 to 58 501 321)	-2121 (-91 991 to 94 189)	40 178 (-43 635 to 134 436)
QALYs	3 730 012 (3 724 192 to 3 736 374)	3 730 977 (3 725 733 to 3 737 451)	3 732 301 (3 725 901 to 3 739 595)	965 (-3551 to 5341)	1324 (-2602 to 5669)
Incremental cost-effectiveness ratio, \$/QALY	NA	NA	NA	Cost saving	30 348
Secondary outcomes					
Proportion of patients with intensified treatment, %	50.7	57.3	64.1	6.60	6.80
Life-years	7 036 648 (6 724 973 to 7 152 202)	7 038 509 (6 726 723 to 7 153 892)	7 041 702 (6 729 712 to 7 158 154)	1861 (-9701 to 12 424)	3193 (-6770 to 14 072)
Total ASCVD events ^c	129 811 (123 505 to 133 965)	129 194 (122 891 to 133 243)	128 176 (121 475 to 132 471)	-617 (-1442 to 245)	-1018 (-1974 to -6)
Incident ASCVD events ^d	82 165 (78 556 to 84 630)	81 789 (78 227 to 84 208)	81 133 (77 310 to 83 701)	-377 (-837 to 65)	-655 (-1192 to -57)
Total CHD events	69 480 (65 953 to 71 848)	69 084 (65 560 to 71 521)	68 446 (64 661 to 70 994)	-396 (-943 to 164)	-637 (-1284 to 59)
Total strokes	60 331 (57 508 to 62 172)	60 110 (57 240 to 61 837)	59 730 (56 840 to 61 555)	-221 (-810 to 334)	-380 (-1020 to 249)

Abbreviations: ApoB, apolipoprotein B; ASCVD, atherosclerotic cardiovascular disease; CHD, coronary heart disease; HDL-C, high-density lipoprotein cholesterol; LDL-C, low-density lipoprotein cholesterol; NA, not applicable; QALY, quality-adjusted life-year; UI, uncertainty interval.

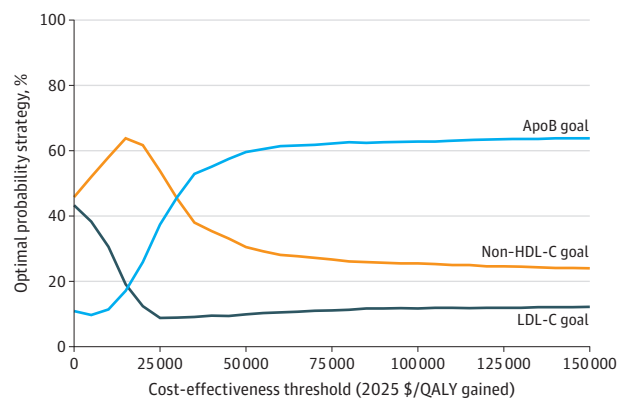
^a Results for statin-eligible individuals at baseline.

^b Uncertainty interval is derived from 2.5th and 97.5th percentile values from 1000 probabilistic model iterations.

^c All projected CHD and stroke events.

^d All projected primary CHD and stroke events.

Figure 1. Line Graph Showing Cost-Effectiveness Acceptability Curve



Probability each strategy is cost-effective (most QALYs with incremental cost-effectiveness ratio less than cost-effectiveness threshold). Derived from 1000 probabilistic simulations of 250 000 statin-eligible patients. ApoB indicates apolipoprotein B; HDL-C, high-density lipoprotein cholesterol; LDL-C, low-density lipoprotein cholesterol; and QALY, quality-adjusted life-year.

Sensitivity Analyses

In almost all sensitivity analyses, intensification based on apoB goals produced the most QALYs of all intensification strategies and was high value (eFigures 7 and 8 in Supplement 1). One-way sensitivity analyses showed that relative risk of stroke per reduction in apoB of 10 mg/dL and the apoB-lowering effectiveness of various lipid-lowering therapies had the largest effect on cost-effectiveness when either usual care with non-HDL-C goal or non-HDL-C goal with apoB goal was compared.

When the effect of using LDL-C, non-HDL-C, and apoB goals that intensified treatment for the same number of people was examined, apoB-guided intensification consistently prevented the most ASCVD events and produced the most QALYs (Figure 2; eTable 11 in Supplement 1). Although apoB-guided intensification was cost-effective compared with LDL-C and non-HDL-C, intensifying treatment for all individuals was the most cost-effective treatment option (ICER, \$24 600/QALY), reflecting the highly cost-effective nature of statin therapy.

Discussion

In accordance with a computer simulation model, using apoB goals to guide LLT intensification was estimated to be cost-effective compared with the current standard LDL-C goal and non-HDL-C goals. Of the LLT goals evaluated, an apoB goal was estimated to produce the largest reduction in ASCVD events and the largest gain in population health. To ensure a robust analysis, a probabilistic analysis was conducted, with 1000 probabilistic iterations of the model. Probabilistic results indicate only a 10% probability that the current standard of care, an LDL-C goal to intensify LLT, is the most cost-effective option. Conversely, there is a 65% probability that an apoB goal would be optimal. These findings indicate the superiority of apoB while appropriately accounting for uncertainty.

Prior work has consistently demonstrated that indices of particle number, particularly apoB, are superior markers of on-treatment risk for ASCVD events.⁶⁻⁸ Taking a similar analytic approach, 1 prior analysis demonstrated cost-effectiveness of using a nuclear magnetic resonance spectroscopy-derived LDL particle number over LDL-C when statin intensity was

Table 3. Disaggregated Lifetime Costs for LDL-C-, Non-HDL-C-, and ApoB-Guided Treatment Intensification^a

Costs in millions of dollars, 2025	Mean (95% UI) ^b				
	LDL-C goal (100 mg/dL), usual care	Non-HDL-C goal (118 mg/dL)	ApoB goal (78.7 mg/dL)	Non-HDL-C goal vs LDL-C goal	ApoB goal vs non-HDL-C goal
Total health care ^c	58 062 (57 648 to 58 471)	58 060 (57 645 to 58 461)	58 100 (57 682 to 58 501)	-2.12 (-92.0 to 94.2)	40.2 (-43.6 to 134)
Lipid-lowering therapy	1003 (788 to 1294)	1015 (799 to 1307)	1070 (861 to 1367)	12.4 (8.9 to 16.1)	55.0 (35.3 to 75.8)
Screening	10.7	10.7	15.8	No difference	5.13
ASCVD health care ^d	4914 (4554 to 5522)	4885 (4536 to 5491)	4846 (4494 to 5462)	-28.6 (-63.5 to 12.0)	-38.8 (-85.1 to 6.78)
Non-ASCVD health care ^e	52 135 (51 381 to 52 420)	52 149 (51 397 to 52 430)	52 168 (51 401 to 52 450)	14.1 (-69.1 to 96.9)	18.9 (-55.7 to 97.8)

Abbreviations: ApoB, apolipoprotein B; ASCVD, atherosclerotic cardiovascular disease; HDL-C, high-density lipoprotein cholesterol; LDL-C, low-density lipoprotein cholesterol; UI, uncertainty interval.

^a N = 250 000 simulated individuals, statin-eligible at baseline.

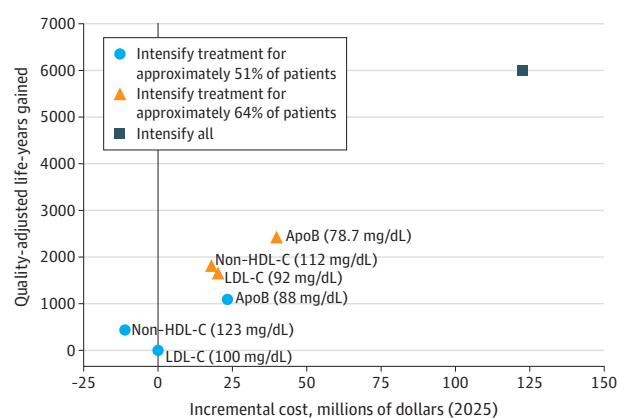
^b Uncertainty interval is derived from 2.5th and 97.5th percentile values from 1000 probabilistic model iterations.

^c Sum of lipid-lowering therapy, screening, ASCVD, and non-ASCVD health care costs across lifetime horizon of cohort.

^d All long-term and acute health care costs associated with ASCVD events minus lipid-lowering therapy.

^e All health care costs minus long-term and acute ASCVD costs, screening costs, and lipid-lowering therapy.

Figure 2. Scatterplot Showing Cost-Effectiveness of Alternative LDL-C (Low-Density Lipoprotein Cholesterol), Non-HDL-C (High-Density Lipoprotein Cholesterol), and ApoB (Apolipoprotein B) Goals



Incremental costs and quality-adjusted life-years gained from intensifying lipid-lowering therapy under different on-treatment lipoprotein thresholds in a deterministic microsimulation analysis of 250 000 individuals. Light blue points represent intensification of treatment for approximately 51% of the population; orange, for approximately 64%; and dark blue, for all individuals.

titrated.²³ However, to our knowledge, no prior study has explored the cost-effectiveness of apoB compared with LDL-C and non-HDL-C. Results of this study indicate that, when treatment for a fixed number of individuals is intensified, apoB consistently produces greater health benefits than LDL-C and non-HDL-C goals. Although an intensify-all strategy would be the most cost-effective option, the clinical acceptability of this strategy is unclear.

This study's findings are logical when viewed in the context of the known biology of apoB and ASCVD. Apolipoprotein B serum concentrations reflect the total number of atherogenic lipoproteins present in the blood. Because the cholesterol and triglyceride content of these particles can vary widely within and between individuals, standard measures of cholesterol and triglyceride mass do not precisely quantify

global atherogenic particle burden.² All apoB-containing lipoproteins can gain entry and be retained in the subendothelial space, the central process underlying atherosclerotic plaque initiation, propagation, instability, and eventual rupture for some individuals.²⁴ Furthermore, reducing the burden of these particles reduces plaque volume, increases stability, and reduces downstream ASCVD risk.²⁵ Thus, reductions in apoB concentration predict ASCVD events more accurately than measures of cholesterol and triglyceride concentration in the blood.

Although the superiority of apoB compared with LDL-C and non-HDL-C to predict risk and guide LLT intensity has been well established,^{2,3} there have been concerns about adopting apoB as the preeminent lipid marker for clinical care, including those related to testing costs. As costs of ASCVD in the US increase rapidly and effective yet costly therapies to reduce ASCVD risk emerge, it is crucial to give proper attention to the optimal therapeutic target for lipid management. This study's model assumes that for each treatment strategy, both an apoB and lipid panel are obtained.

Results of this study indicate that apoB-guided treatment intensification reduced ASCVD-related health care costs but led to a net increase in health care costs. The model uses a lifetime horizon, from treatment initiation to death. Increased costs in the apoB-guided intensification group reflect the fact that identifying high-benefit patients more accurately can extend lifetime health care costs by promoting increased life-years. Although there is an added cost to obtain an apoB assay compared with a lipid panel, the results suggest that the increased cost of the assay itself plays a minimal role in the cost-effectiveness of using an apoB goal compared with an LDL-C or non-HDL-C goal. Additionally, if apoB is a superior decisional marker, then it could be argued that one may forgo concurrently obtaining a lipid panel and that universal uptake of an apoB target could also drive down laboratory costs because of economy of scale.

The incremental QALY estimates in this analysis reflect uncertainty from treatment effects, adverse events, utility decrements, ASCVD event costs, and patient-level variation. Although the UI for this estimate crosses zero, the probability that

an apoB goal is cost-effective is best represented by the proportion of probabilistic iterations of the model falling below the willingness-to-pay threshold.²² In the analysis, these results consistently favor apoB-guided intensification, supporting the robustness of the conclusions.

Limitations

In the analysis, assumptions were made related to the cost of assays and statins and the benefit of lipid-lowering therapies, as outlined in the Methods. Although these assumptions were informed by the best available evidence, the base case results are contingent on the validity of these assumptions. The study undertook extensive 1-way sensitivity and scenario analyses. Results from these analyses suggest that the findings remain valid across a wide range of plausible modeling assumptions. In addition, the study was conducted from a health sector per-

spective and omits broader societal benefits that could be achieved through reduced ASCVD events (eg, reduced workplace absenteeism). Future analyses could incorporate a societal perspective, although it remains unclear whether health care payers should pay for nonhealth benefits.⁶

Conclusions

It has been consistently demonstrated that indices of particle number, particularly apoB, are superior markers of on-treatment risk for ASCVD events compared with LDL-C or non-HDL-C.^{2,3} This computer simulation study supports the use of apoB as a cost-effective marker to guide LLT in primary prevention clinical scenarios for health care payers in the United States.

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